

Autonomous Vehicle Simulation (AVS) Laboratory

Basilisk Technical Memorandum

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ALGORITHMS TO MAP DESIRED TORQUE VECTOR ONTO A SET OF THRUSTERS

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Status: Ready
Scope/Contents
This module takes a commanded attitude control torque vector and determines a set of desired thruster force values to implement this torque. It is assumed that the nominal thruster configuration is such that pure torque solutions are possible. The module supports both on- and off-pulsing solutions, including cases where the thruster solutions are saturated due to a large commanded attitude control torque.

Rev	Change Description	By	Date
1.0	Initial Documentation	H. Schaub	2019-02-09
1.1	Updated the thruster force evaluation to account for center of mass offsets	H. Schaub	

1.2	Updated the figure and the $[C]$ matrix notation	H. Schaub	
1.3	The thruster mapping logic has changed, and this documentation now reflects what the new algorithm does.	H. Schaub	
1.4	Discuss the new module feature of scaling the thruster force solution during periods where the thrusters are saturated.	H. Schaub	
2.0	Update document to adhere to new documentation format	J. Martin	2019-04
2.1	Compute the torque angle error with the reduced torque in body frame components	C. Cena	2026-10

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1 Module Description

1.1 Introduction

1.1.1 Torque Control Axes

This technical note describes a general algorithm that maps an input ADCS control torque $\boldsymbol{L}_{\text{cmd}}$ onto force commands for a cluster of thrusters. In the on-pulsing formulation below, the allocator target is $\boldsymbol{L}_r = \boldsymbol{L}_{\text{cmd}}$. Section 1.4 defines the nominal-thrust compensation used to obtain $\boldsymbol{L}_r$ for off-pulsing. Let $\hat{\boldsymbol{c}}_j$ be the axis about which the thrusters are to produce the desired torque. The matrix of N_c thruster axes rows is then given by

$$[C] = \begin{bmatrix} \hat{\boldsymbol{c}}_1 \\ \vdots \\ \hat{\boldsymbol{c}}_{N_c} \end{bmatrix} \tag{1}$$

The module can accept up to 3 orthogonal control axis $\hat{\boldsymbol{c}}_j$. Let $\bar{\boldsymbol{L}}_r$ be the three-dimensional reduced set of $\boldsymbol{L}_r$ onto the set of control axes written in the body-frame, given by:

$$^{\mathcal{B}}\bar{\boldsymbol{L}}_r = [C]^T[C]^{\mathcal{B}}\boldsymbol{L}_r \tag{2}$$

The goal of the thruster mapping strategy is to find a set of $\boldsymbol{F}$ thruster forces that yield $\bar{\boldsymbol{L}}_r$.

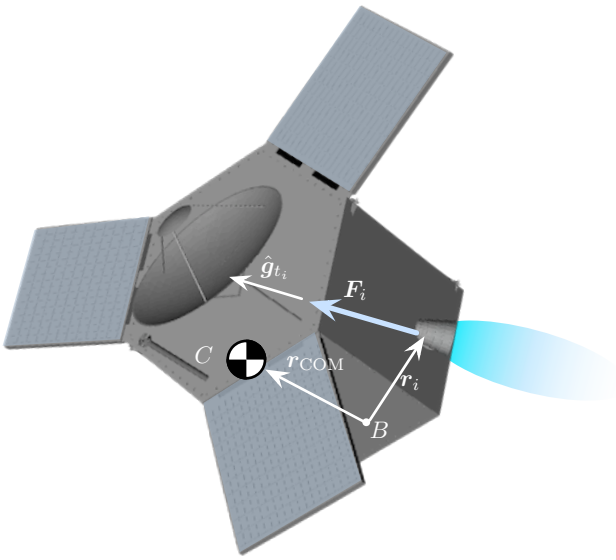


Fig. 1: Illustration of the Spacecraft Thruster Notation

1.1.2 Thruster to Torque Mapping Matrix

The ith thruster location relative to the spacecraft point B is given by $\boldsymbol{r}_i$ as illustrated in Figure 1. The unit direction vector of the thruster force is $\hat{\boldsymbol{g}}_{t_i}$, while the thruster force is given by

$$\boldsymbol{F}_i = F_i \hat{\boldsymbol{g}}_{t_i} \tag{3}$$

The toque vector produced by each thruster about the body fixed point C is thus

$$\boldsymbol{\tau}_i = (\boldsymbol{r}_i - \boldsymbol{r}_{\text{COM}}) \times F_i \hat{\boldsymbol{g}}_{t_i} \tag{4}$$

The total torque onto the spacecraft, due to a cluster of N thrusters, is

$$\tau_j = \sum_{i=1}^{N} \boldsymbol{\tau}_i = \sum_{i=1}^{N} ((\boldsymbol{r}_i - \boldsymbol{r}_{\text{COM}}) \times \hat{\boldsymbol{g}}_{t_i}) F_i = \sum_{i=1}^{N} \boldsymbol{d}_i F_i \tag{5}$$

where

$$\boldsymbol{d}_i = (\boldsymbol{r}_i - \boldsymbol{r}_{\text{COM}}) \times \hat{\boldsymbol{g}}_{t_i} \tag{6}$$

In matrix form, the net spacecraft torque is written compactly as

$$\boldsymbol{\tau} = \begin{bmatrix} \boldsymbol{d}_1 \cdots \boldsymbol{d}_N \end{bmatrix} \begin{bmatrix} F_1 \\ \vdots \\ F_N \end{bmatrix} = [D]\boldsymbol{F} \tag{7}$$

where $[D]$ is a $3 \times N$ matrix that maps the thruster forces F_i to the spacecraft torque $\boldsymbol{\tau}$.

1.2 ACS Thruster Force Algorithm for a Thruster Configuration with Full Torque Controllability

Here a thruster configuration is assumed that can produce pure torque-couples without exerting a net force onto the spacecraft. The thrusters force values F_i must be strictly non-negative (i.e. either 0 or

positive). Note that in this configuration having all thrusters on will produce zero net force and torque onto the spacecraft. Thus the $F_i = F_j$ solution is in the nullspace of the mapping in Eq. (7).

The goal of the thruster force algorithm is to determine a set of thruster forces $\boldsymbol{F}$ such that the net torque $\boldsymbol{\tau}$ onto the spacecraft is

$$^{\mathcal{B}}\boldsymbol{\tau} = {}^{\mathcal{B}}\bar{\boldsymbol{L}}_r = [D]^{\mathcal{B}}\boldsymbol{F} \tag{8}$$

without bleeding torque onto the un-controlled axes. The first step is to perform a standard minimum norm inverse solution using

$$^{\mathcal{B}}\boldsymbol{F} = [D]^T([D][D]^T)^{-1\mathcal{B}}\bar{\boldsymbol{L}}_r \tag{9}$$

The 3×3 matrix $[D][D]^T$ is full rank and thus invertible with the assumption that this RCS configuration has a full 3D torque controllability. This set of thruster forces will contain F_i values that are both positive and negative. Next, to achieve strictly non-negative values, the minimum F_i value is determined and subtracted from all N force values.

$$\boldsymbol{F} \leftarrow \boldsymbol{F} - \min(\boldsymbol{F}) \tag{10}$$

The resulting set of F_i forces will produce the desired control torque $\bar{\boldsymbol{L}}_r$ and achieve a net zero force onto the spacecraft. The latter results is due to the assumption off an ACS thruster configuration that can produce pure moment couples.

1.2.1 ACS Thruster Force Algorithm for a Thruster Configuration with Partial Torque Controllability

In the case a control frame is not fully defined (i.e. only 1 or 2 control axes are explicitly specified), the math above needs to be slightly modified to compute the torques in the desired control space. Specifically, the mapping matrix $[D]$ and requested torque vector $\mathbf{L_r}$ must be projected into, and operated within, the control subspace. As such,

$$[C] = \begin{bmatrix} \hat{\boldsymbol{c}}_1 \\ \vdots \\ \hat{\mathbf{0}}_{N_c} \end{bmatrix} \tag{11}$$

$$^{\mathcal{C}}\bar{\boldsymbol{L}}_r = [C]^{\mathcal{B}}\boldsymbol{L}_r \tag{12}$$

$$[CD] = [C][D] \tag{13}$$

$$^{\mathcal{B}}\boldsymbol{F} = [CD]^T([CD][CD]^T)^{-1\mathcal{C}}\bar{\boldsymbol{L}}_r \tag{14}$$

Note: $([CD][CD]^T)^{-1}$ is not an invertable matrix if one or more of the control axes $\hat{\boldsymbol{c}}$ is a $\mathbf{0}$ vector. To circumvent this problem, two mathematical reconciliations must be made. First, when $\boldsymbol{L}_r$ is projected onto the control axes, $[C]^{\mathcal{B}}\boldsymbol{L}_r$, the component projected onto the $\mathbf{0}$ axis will be removed. I.e. if

$$[C] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

and

$$^{\mathcal{B}}\boldsymbol{L}_r = (1.0, -0.5, 0.7)^T$$

then

$$^{\mathcal{C}}\boldsymbol{L}_r = (1.0, 0.7, 0.0)^T$$

and

$$[CD] = \begin{pmatrix} CD_{11} & CD_{12} & \dots & CD_{1N} \\ CD_{21} & CD_{22} & \dots & CD_{2N} \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

Second, the $[M] = ([CD][CD]^T)$ matrix must be set to identity, and only the upper block entries in $[M]$ will be computed (2x2 or 1x1 depending on 2 or 1 control axes respectively) . So for the 2-axis control scheme,

$$[M] = \begin{pmatrix} M_{11} & M_{12} & 0 \\ M_{21} & M_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Allowing $[M]$ to be inverted. This assertion can be made because when mapping the control torques onto the control axes, the $[M]_{33} = 1$ component is multiplied by the zero in ${}^{\mathcal{C}}\boldsymbol{L}_r$. Thus:

$${}_B \begin{pmatrix} F_1 \\ F_2 \\ \vdots \\ F_N \end{pmatrix} = \begin{pmatrix} CD_{11} & CD_{21} & 0 \\ CD_{12} & CD_{22} & 0 \\ \vdots & \vdots & \vdots \\ CD_{1N} & CD_{2N} & 0 \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} & 0 \\ M_{21} & M_{22} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \\ 0 \end{pmatrix} \tag{15}$$

such that ${}^{\mathcal{C}}L_3$ will never be mapped onto the body frame force components ${}^{\mathcal{B}}F$.

1.3 2-Stage Minimum Norm ACS Thruster Mapping Algorithm

To increase the robustness of the sign-constrained minimum norm thruster force solution 2nd stage is included. This is helpful in configurations where the number of available ACS thrusters is not equal to the number of installed thrusters. This simulates scenarios where some thrusters are now offline. To engage this 2nd loop the module flag `use2ndLoop` must be set to 1. The default value is zero where the 2nd loop is not engaged. The minimum norm solution from the earlier solution is first evaluated, and then shifted by subtracting the minimum F_i value.

Each thruster can only produce a positive force. With off-pulsing, the nominal thrust force plus the negative correction must still yield a non-negative thrust force. The module parameter `thrForceSign` is either $+1$ or -1 to account for the desired force sign. The value of this parameter is represented through s_F. With the ACS thruster configuration this value would always be $+1$.

We assume that $\boldsymbol{F}$ elements only contain forces that are either zero or values with the desired sign. Assume there are M force values in $\boldsymbol{F}$ with a sign that matches s_F. The locations of these values is provided in the N-dimensional array $\boldsymbol{t}_{\text{used}}$ which contains either 0 or 1 values. For example, consider $N = 8$ and only thrusters 2 and 6 produce zero forces. In this case we find

$$\boldsymbol{t}_{\text{used}} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{bmatrix} \tag{16}$$

This reduces the thruster force search to a subset of M thrusters. Let $\bar{\boldsymbol{F}}_j$ be a $M \times 1$ matrix of to be determined thruster forces. The corresponding $3 \times M$ mapping matrix $[\bar{D}]$ that projects $\bar{\boldsymbol{F}}$ onto a net body torque about point B is defined as:

$$[\bar{D}] = \begin{bmatrix} \bar{\boldsymbol{d}}_1 & \cdots & \bar{\boldsymbol{d}}_M \end{bmatrix} \tag{17}$$

with

$$\bar{\boldsymbol{d}}_i = (\boldsymbol{r}_i - \boldsymbol{r}_{\text{COM}}) \times \hat{\boldsymbol{g}}_i \tag{18}$$

The net torque due to $\bar{\boldsymbol{F}}$ is

$${}^{\mathcal{B}}\bar{\boldsymbol{\tau}} = [\bar{D}]^{\mathcal{B}}\bar{\boldsymbol{F}} \tag{19}$$

A modified set of thruster force solutions $\bar{\boldsymbol{F}}$ to generate the desired torque $\bar{\boldsymbol{L}}_r$ is found through a second minimum norm operation:

$$^{\mathcal{B}}\bar{\boldsymbol{F}} = [\bar{D}]^T([\bar{D}][\bar{D}]^T)^{-1\mathcal{B}}\bar{\boldsymbol{L}}_r \tag{20}$$

The next step is to sum the individual $\bar{\boldsymbol{F}}$ thruster solutions to the yield the net set of thruster forces required to produce $\bar{\boldsymbol{L}}_r$. This is done using the $\boldsymbol{t}_{\text{used}}$ matrix to determine which thrusters have non-zero contributions. The final step is to evaluate the minimum F_i force again and subtract this from all thruster force values.

1.4 DV Thruster Firing Strategy

With a DV thruster configuration the thruster force axes are parallel. As such, this configuration cannot produce a torque along the DV thrust axis. A slightly modified version of the above algorithm is used in this case. With the DV configuration the attitude along the axes orthogonal to the thrust vector are controlled via off-pulsing. As such, the thruster firing mapping must produce negative F_i values and the `thrForceSign` sign must be set to -1. With this `thrForceSign` value the flag `use2ndLoop` is ignored and the second loop is always engaged.

Here $\boldsymbol{F}$ represents signed changes from the nominal all-on force vector $\boldsymbol{F}_{\text{max}}$. A zero command leaves a thruster fully on, while $F_i = -F_{\text{max},i}$ requests that it remain off for the control period. Before either minimum norm solve, the module compensates for the nominal torque:

$$\boldsymbol{L}_r = \boldsymbol{L}_{\text{cmd}} - [D]\boldsymbol{F}_{\text{max}}, \qquad ^{\mathcal{B}}\bar{\boldsymbol{L}}_r = [C]^T[C]\,^{\mathcal{B}}\boldsymbol{L}_r.$$

Thus $[D]\boldsymbol{F}$ is the change from nominal torque. When the commands are within their limits, the ideal average net thruster torque is $[D](\boldsymbol{F}_{\text{max}} + \boldsymbol{F})$. Nominal torque need not be zero, for example when the spacecraft center of mass is offset from the nominal thrust line.

The modified algorithm still evaluates the first minimum norm inverse, but does not subtract out the $\min(\boldsymbol{F})$ value. Rather, the 2nd stage is used to determine which DV thrusters produce the torque with a negative torque value to generate the corresponding $[\bar{D}]$ matrix. After performing the 2nd minimum norm inverse with $[\bar{D}]$ the subtraction of $\min(\boldsymbol{F})$ is not performed.

1.5 Saturating the Thruster Capability

The thruster force solution $\boldsymbol{F}$ is implemented using pulse-width modulation. For on-pulsing, F_i is a nonnegative average force; for off-pulsing, it is a nonpositive change from the nominal average force $F_{\text{max},i}$. A command with $|F_i| > F_{\text{max},i}$ cannot be realized over the control period. To evaluate the torque direction under saturation, the module clips each signed command to its individual limit:

$$F_{\text{clip},i} = \min\left(F_{\text{max},i}, \max\left(-F_{\text{max},i}, F_i\right)\right), \qquad \boldsymbol{\tau} = [D]\boldsymbol{F}_{\text{clip}}.$$

Without saturation, $\boldsymbol{\tau} = [D]\boldsymbol{F}$. Clipping can change both the magnitude and direction of this torque contribution. For on-pulsing, it is the ideal average net thruster torque. For off-pulsing, the net torque also includes the nominal contribution:

$$\boldsymbol{\tau}_{\text{net}} = [D](\boldsymbol{F}_{\text{max}} + \boldsymbol{F}_{\text{clip}}).$$

The module's angle compares the signed torque contribution $\boldsymbol{\tau}$ with the reduced allocator target $\bar{\boldsymbol{L}}_r$, which includes the nominal-thrust compensation in off-pulsing mode. It does not compare $\boldsymbol{\tau}_{\text{net}}$ directly with the input request $\boldsymbol{L}_{\text{cmd}}$.

$$\theta_{\boldsymbol{\tau}/\bar{\boldsymbol{L}}_r} = \arccos\left(\frac{\boldsymbol{\tau}\cdot\bar{\boldsymbol{L}}_r}{|\boldsymbol{\tau}||\bar{\boldsymbol{L}}_r|}\right) \tag{21}$$

In Eq. (21), both $\boldsymbol{\tau} = [D]\boldsymbol{F}_{\text{clip}}$ and $\bar{\boldsymbol{L}}_r$ are expressed in body frame components. Thus ${}^{\mathcal{B}}\bar{\boldsymbol{L}}_r = [C]^T[C]\,{}^{\mathcal{B}}\boldsymbol{L}_r$ is used instead of the control frame components ${}^{\mathcal{C}}\bar{\boldsymbol{L}}_r$ in the force solution of Eq. (14). The angle is invariant under reordering the control axes or selecting another orthonormal basis for the same controlled subspace. Changing that subspace can change the angle and force solution.

In the 8-thruster configuration in section 4.0.1, some thrusters support multiple control axes, so saturation can change the torque direction substantially. Uniformly scaling the signed commands until every $|F_i| \leqslant F_{\text{max},i}$ removes clipping and preserves the direction of their unscaled torque contribution $[D]\boldsymbol{F}$. This contribution aligns with $\bar{\boldsymbol{L}}_r$ only if it was already aligned before scaling. In particular, torque outside the controlled subspace is also scaled and can leave a nonzero angle error. This scaling rule applies to both on- and off-pulsing solutions. In off-pulsing mode, the net torque also includes the unchanged nominal contribution $[D]\boldsymbol{F}_{\text{max}}$, so its direction need not be preserved by scaling the commands.

The parameter `angErrThresh` sets the angular threshold for scaling. When the angle in Eq. (21) exceeds this threshold and at least one commanded force exceeds its limit, all commanded forces are scaled by the same positive factor such that $|F_i| \leqslant F_{\text{max},i}$. The module then recomputes `outTorqAngErr`. At or below the angle threshold, the force commands can remain above their limits even though the angle calculation uses clipped forces.

The default value of `angErrThresh` is 0 rad. If a control torque misalignment of 10° is acceptable, set `angErrThresh` to $10\pi/180$ rad.

To disable scaling, set `angErrThresh` to a value larger than π rad (180°). The torque angle cannot exceed this value, so the scaling condition is never reached.

2 Module Functions

This module has the following functions:

- **Read in the thruster and spacecraft configuration**: When reset the module reads in the thruster locations, force directions and maximum thrust values, as well as the spacecraft center of mass. The control axes $\hat{\boldsymbol{c}}_j$ are normalized, and a warning is logged if any two control axes are not orthogonal. On every update the module reads the current spacecraft center of mass and recomputes the thruster moment arms.

- **Compute the thruster force solution**: With each update call the module forms the allocator target $\boldsymbol{L}_r$. For on-pulsing, this is the input torque $\boldsymbol{L}_{\text{cmd}}$; for off-pulsing, the nominal torque $[D]\boldsymbol{F}_{\text{max}}$ is subtracted first. The module reduces this target onto the control axes and computes a minimum norm solution for the signed force commands $\boldsymbol{F}$. The solution can also produce torque outside the controlled subspace. If `use2ndLoop` is set or off-pulsing is used (`thrForceSign` $= -1$), the 2-stage solution is used to enforce the desired force sign.

- **Evaluate the torque angle error**: The angle $\theta_{\boldsymbol{\tau}/\bar{\boldsymbol{L}}_r}$ between the saturation-clipped torque contribution $\boldsymbol{\tau} = [D]\boldsymbol{F}_{\text{clip}}$ and the reduced allocator target $\bar{\boldsymbol{L}}_r$ is evaluated in body frame components, and is stored in `outTorqAngErr`. In off-pulsing mode, this compares the change from nominal torque with the compensated target.

- **Scale the thruster forces if saturated**: If this angle is larger than `angErrThresh`, the thruster forces are scaled such that no thruster exceeds its maximum thrust.

- **Output the thruster force commands**: The resulting thruster forces are written to the thruster force output message.

3 Module Assumptions and Limitations

The module assumes all thrusters in the configuration message are operating and available. Thruster locations, force directions, and maximum thrust values are cached until the next module reset. The spacecraft center of mass is read from `vehConfigInMsg` on every update, so changes to `CoM_B` do not require a reset. Up to three control axes are supported, and these must be orthogonal for the reduced torque $\bar{\boldsymbol{L}}_r$ to be valid.

4 Test Description and Success Criteria

The module test `test_module` is run with the following parameters:

1. useDVThruster: Determines if using a DV thruster configuration or a RCS thruster configuration
2. useCOMOffset: Offset the COM from the control axes.
3. dropThruster: Removes 0, 1 or 2 thrusters from the DV or RCS configuration. When thrusters are dropped the 2nd least squares fitting loop `use2ndLoop` is used.
4. numControlAxis: Number of controlled axes (1, 2 or 3). The DV configuration is only evaluated for 1 and 2 control axes.
5. saturateThrusters: Changes the requested torque and the angle tolerance which triggers scaling behavior.

The control axes of `test_module` are the leading `numControlAxis` body axes.

The test `test_moduleAxesOrder` reruns a representative subset of these cases (no COM offset, 0 or 1 dropped thrusters, `saturateThrusters` of 0 or 2, all values of `useDVThruster` and `numControlAxis`) with the additional parameter

1. axesOrder: Uses the same leading body axes in reversed order, or an orthonormal rotation of these axes, to span the same controlled subspace. The thruster forces and the torque angle error `outTorqAngErr` must not depend on this choice. For a single control axis the reversed set equals the leading set and the rotated set only negates the axis; these cases are kept for clarity.

The test results below are reported for the leading body axes.

4.0.1 ACS Thruster Configuration

To illustrate the performance of this algorithm, the following simulation setup is used. Let the ACS system have a total of $N = 8$ thrusters with the following body-fixed locations:

$$\begin{aligned}
\boldsymbol{r}_1 &= \left[+1.125, 0.0, +0.75\right]^T \text{ m} & \boldsymbol{r}_2 &= \left[-1.125, 0.0, +0.75\right]^T \text{ m} \\
\boldsymbol{r}_3 &= \left[-1.125, 0.0, +0.75\right]^T \text{ m} & \boldsymbol{r}_4 &= \left[+1.125, 0.0, +0.75\right]^T \text{ m} \\
\boldsymbol{r}_5 &= \left[+1.125, 0.0, -0.75\right]^T \text{ m} & \boldsymbol{r}_6 &= \left[-1.125, 0.0, -0.75\right]^T \text{ m} \\
\boldsymbol{r}_7 &= \left[-1.125, 0.0, -0.75\right]^T \text{ m} & \boldsymbol{r}_8 &= \left[+1.125, 0.0, -0.75\right]^T \text{ m}
\end{aligned}$$

with the force unit direction vectors:

$$\begin{aligned}
\boldsymbol{g}_{t_1} &= \left[+0.707107, +0.707107, 0\right]^T & \boldsymbol{g}_{t_2} &= \left[-0.707107, +0.707107, 0\right]^T \\
\boldsymbol{g}_{t_3} &= \left[-0.707107, -0.707107, 0\right]^T & \boldsymbol{g}_{t_4} &= \left[+0.707107, -0.707107, 0\right]^T \\
\boldsymbol{g}_{t_5} &= \left[+0.707107, +0.707107, 0\right]^T & \boldsymbol{g}_{t_6} &= \left[-0.707107, +0.707107, 0\right]^T \\
\boldsymbol{g}_{t_7} &= \left[-0.707107, -0.707107, 0\right]^T & \boldsymbol{g}_{t_8} &= \left[+0.707107, -0.707107, 0\right]^T
\end{aligned}$$

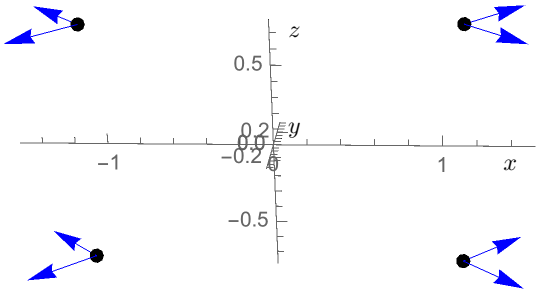


Fig. 2: Illustration of an 8-thruster ACS configuration

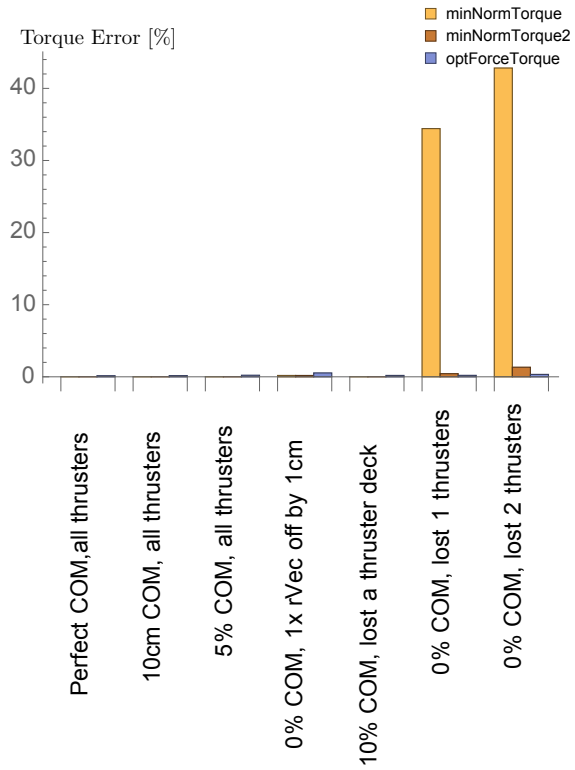
The resulting configuration is illustrated in Figure 2. In this setup the center of mass position vector is set to $\boldsymbol{r}_{\text{COM}} = (0, 0, 0)^T$ m. The following plots show the thruster firing performance by considering a rang of scenarios. The first case assumes all thrusters are available, and the $\boldsymbol{r}_{\text{COM}}$ vector is known perfectly. The second and third case also have all thrusters available, but the $\boldsymbol{r}_{\text{COM}}$ vector knowledge is off by 10cm or 5cm in the z direction. The 4th case has the perfect $\boldsymbol{r}_{\text{COM}}$ vector, but the first thruster location is off by 1cm in the y direction. The 5th case again assumes a 10cm COM offset, but also that only 4/8 thrusters are available. In essence, all the lower or upper thruster have become un-available. The 6th and 7th case assumes perfect COM knowledge, but either the last one or last two thrusters are un-available.

The resulting performance is illustrated in Figure 3. Here 20 random $\boldsymbol{L}_r$ torques are generated, and the mean results are shown. The $[C]$ matrix is set to the identity matrix to control all three axes. The 1-stage minimum norm algorithm is compared to an optimal thruster firing solution which minimizes the net thruster forces used, the net disturbance torque onto the craft, all subject to producing the desired control torque. Regarding generating the desired control torque, only when 1 or 2 thrusters were lost did the algorithm have issues generating the required torque. In all other cases the desired torque was always produced as shown in Figure 3(a). In comparison, the 2-stage minimum norm algorithm performs very well in these latter cases where 1-2 thrusters are lost, yielding only very small torque errors on average.

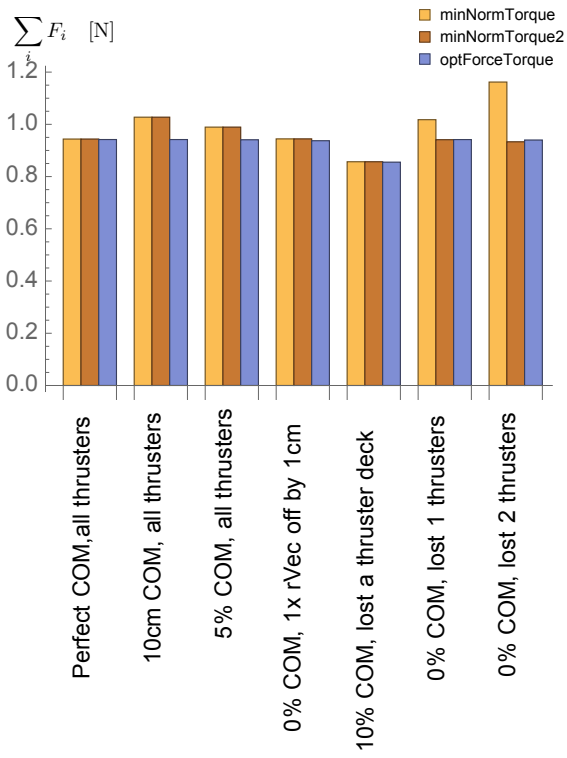
The control effort required to achieve this torque is shown in Figure 3(b). Here the F_i thruster force values are simply summed to have a sense of how much on-time or fuel is required for a given scenario. If all ACS thrusters are operating and the COM is perfectly modeled is the control effort equivalent to the optimal solution. In all other case the effort is about 10-20% larger. in contrast, the 2-stage minimum norm algorithm control efforts in the cases with lost thrusters is very similar to those of the optimal answers. This illustrates the robustness achieved by adding this 2nd stage if some thrusters are lost.

Finally, to see how well this algorithm is able to avoid net disturbance forces onto the spacecraft, the results in Figure 3(c) are shown. Overall the 1-stage algorithm did well, producing 0 net force in the ideal case and the case where the 1st thruster location is different, and matches the optimal answer for the lost thruster deck and lost single thruster case. In the other cases there is some net disturbance force. The 2-stage algorithm only differs here when thrusters are lost. Here the net disturbance torque is higher than the 1-stage solution. However, this is an expected result as the 1-stage solution produces huge torque errors, while the 2-stage solution produces a very different thruster solution which yields very small torque errors.

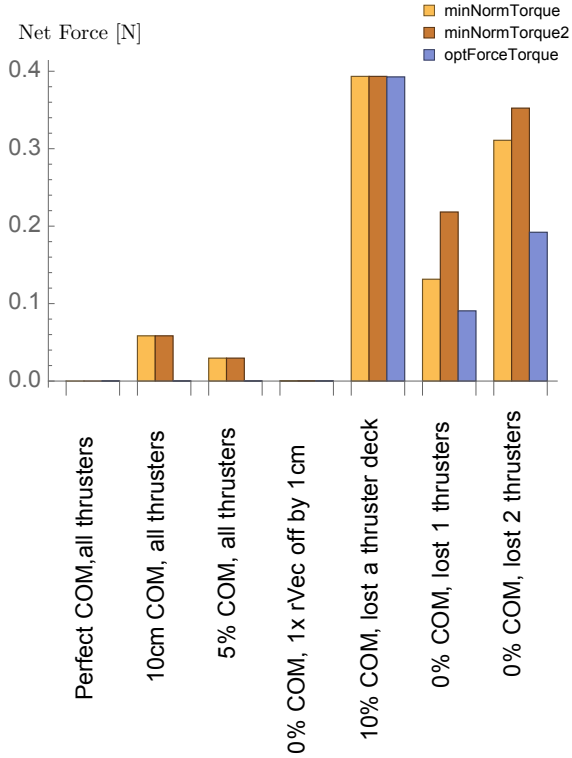
Overall the 2-stage thruster firing performs very well in comparison to the the optimal thruster firing solutions. The optimal solutions are computationally very expensive and not suitable to realtime flight-software implementations.



(a) Percent Torque Error



(b) Thruster Control Implementation Effort



(c) Net Thruster Disturbance Force

Fig. 3: ACS Thruster Mapping Performance Mapping Illustration Comparing the 1-stage and 2-stage minimum norm solution to the thruster mapping of a nonlinear optimization solution.

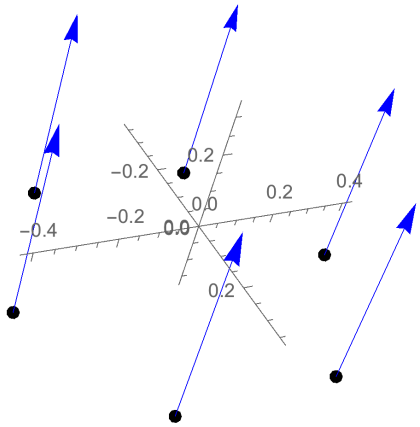


Fig. 4: Illustration of an 6-thruster DV configuration

4.0.2 DV Thruster Configuration

Next consider a DV thruster configuration where the off-thruster-axis control torques are produces through off-pulsing. Let the DV system have a total of $N = 6$ thrusters with the following body-fixed locations:

$$\boldsymbol{r}_1 = \left[0, 0.413, -0.1671\right]^T \text{ m} \qquad \boldsymbol{r}_2 = \left[0.357668, 0.2065, -0.1671\right]^T \text{ m}$$
$$\boldsymbol{r}_3 = \left[0.357668, -0.2065, -0.1671\right]^T \text{ m} \qquad \boldsymbol{r}_4 = \left[0, -0.413, -0.1671\right]^T \text{ m}$$
$$\boldsymbol{r}_5 = \left[-0.357668, -0.2065, -0.1671\right]^T \text{ m} \qquad \boldsymbol{r}_6 = \left[-0.357668, 0.2065, -0.1671\right]^T \text{ m}$$

with the force unit direction vectors given by $\hat{\boldsymbol{g}}_{t_i} = (0, 0, 1)^T$ as illustrated in Figure 4.

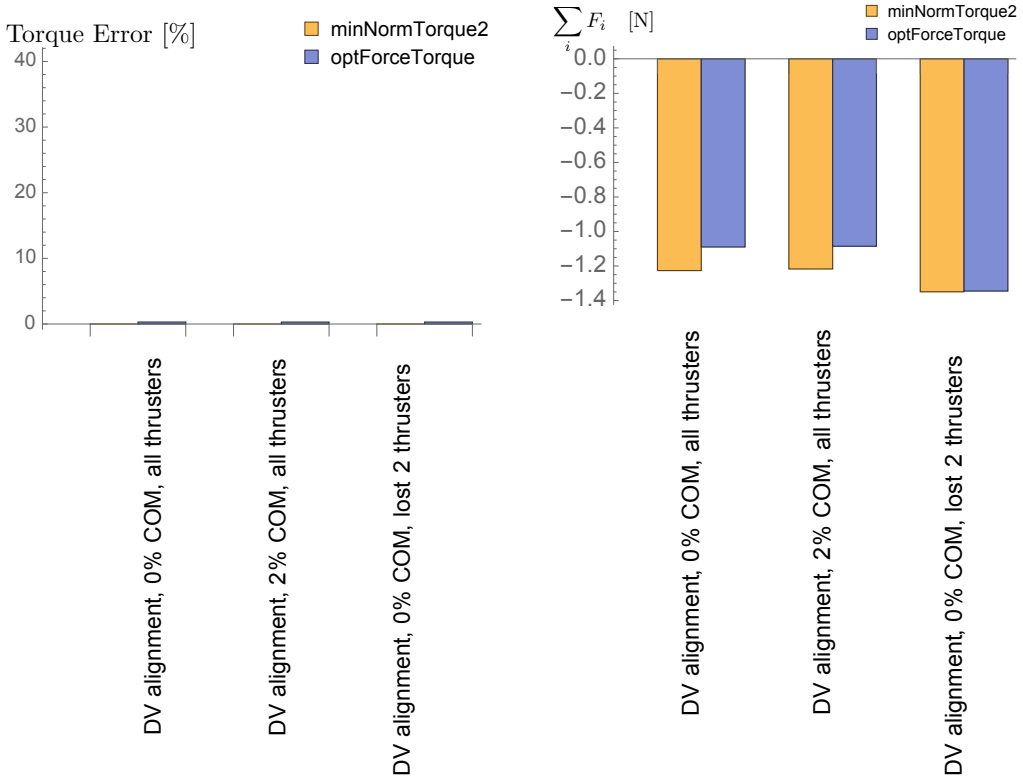
Figure 5 illustrates the DV off-pulsing performance for 3 scenarios. The 1st one has all thrusters available and no COM error. The 2nd case adds 2cm COM offsets to the x and y axes, while the 3rd case has no COM error but lost an opposing set of thrusters. Again 20 random $\boldsymbol{L}_r$ vectors are generated and the mean performance evaluated. As only torques about the x and y axis can be controlled with this DV thruster configuration, the control axis matrix is set to

$$[C] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

The torque implementation errors are compared in Figure 5(a). The 2-stage process outlined above is able to produce the required torques in all cases, illustrating good robustness to COM offsets and loosing 2/6 thrusters. Figure 5(b) illustrates the net off-pulsing effort required to achieve these random control torque vectors. The 2-stage algorithms requires slightly more off-pulsing than the optimal solution. However, the difference isn't very large, especially in comparison to the drastically faster computational evaluation time.

4.0.3 Control Axes Ordering Test

The test `test_controlAxesOrdering` checks the torque angle for several controlled subspaces and verifies that the saturated force solution is unchanged by using different orthonormal bases of the full three-dimensional space. An RCS configuration of 8 thrusters located at the corners of a 2 m cube and firing tangentially is used. The control axis sets considered are the identity $[x, y, z]$, the permuted set $[z, x, y]$, the non-leading set $[x, z]$, the reversed set $[y, x]$, the single axis $[y]$, as well as a rotated orthonormal 3-axis set and a rotated orthonormal 2-axis set. For each case the test checks that



(a) Percent Torque Error (b) Thruster Control Implementation Effort

Fig. 5: DV Thruster Mapping Performance Mapping Illustration Comparing the 1-stage and 2-stage minimum norm solution to the thruster mapping of a nonlinear optimization solution.

- without thruster saturation the produced torque satisfies $[C][D]\boldsymbol{F} = [C]\boldsymbol{L}_r$,
- the output `outTorqAngErr` equals the angle in Eq. (21) between the body frame reduced torque $[C]^T[C]\boldsymbol{L}_r$ and the saturation-clipped torque $[D]\boldsymbol{F}_{\text{clip}}$.

A saturated case is also run with a maximum thrust of 0.2 N and `angErrThresh` $= 10^\circ$. Here the true torque angle error is about 6.5°, while the former control frame comparison yielded about 108°. For the 3-axis sets the test checks that the thruster forces are not scaled and are equal to the forces obtained with the identity control axes.

5 Test Parameters

Table 2: Error tolerance for each test.

Output Value Tested	Tolerated Error
`thrForce`	1e-06

6 Test Results

All of the tests passed:

Table 3: Test results

DVThruster?	useCOMOffset?	dropThruster	numControlAxis	saturateThrusters	Pass/Fail
False	False	0	1	0	PASSED
False	False	0	1	1	PASSED
False	False	0	1	2	PASSED
False	False	0	2	0	PASSED
False	False	0	2	1	PASSED
False	False	0	2	2	PASSED
False	False	0	3	0	PASSED
False	False	0	3	1	PASSED
False	False	0	3	2	PASSED
False	False	1	1	0	PASSED
False	False	1	1	1	PASSED
False	False	1	1	2	PASSED
False	False	1	2	0	PASSED
False	False	1	2	1	PASSED
False	False	1	2	2	PASSED
False	False	1	3	0	PASSED
False	False	1	3	1	PASSED
False	False	1	3	2	PASSED
False	False	2	1	0	PASSED
False	False	2	1	1	PASSED
False	False	2	1	2	PASSED
False	False	2	2	0	PASSED
False	False	2	2	1	PASSED
False	False	2	2	2	PASSED
False	False	2	3	0	PASSED
False	False	2	3	1	PASSED
False	False	2	3	2	PASSED
False	True	0	1	0	PASSED
False	True	0	1	1	PASSED
False	True	0	1	2	PASSED
False	True	0	2	0	PASSED
False	True	0	2	1	PASSED
False	True	0	2	2	PASSED
False	True	0	3	0	PASSED
False	True	0	3	1	PASSED
False	True	0	3	2	PASSED
True	False	0	1	0	PASSED
True	False	0	1	1	PASSED
True	False	0	1	2	PASSED
True	False	0	2	0	PASSED
True	False	0	2	1	PASSED
True	False	0	2	2	PASSED
True	False	1	1	0	PASSED
True	False	1	1	1	PASSED
True	False	1	1	2	PASSED
True	False	1	2	0	PASSED
True	False	1	2	1	PASSED
True	False	1	2	2	PASSED
True	False	2	1	0	PASSED
True	False	2	1	1	PASSED
True	False	2	1	2	PASSED
True	False	2	2	0	PASSED

Table 3: Test results (continued)

DVThruster?	useCOMOffset?	dropThruster	numControlAxis	saturateThrusters	Pass/Fail
True	False	2	2	1	PASSED
True	False	2	2	2	PASSED
True	True	0	1	0	PASSED
True	True	0	1	1	PASSED
True	True	0	1	2	PASSED
True	True	0	2	0	PASSED
True	True	0	2	1	PASSED
True	True	0	2	2	PASSED

7 User Guide

1. ϵ **Parameter**: The minimum norm inverse requires a non-zero determinant value of $[D][D]^T$. For this setup, this matrix is a scalar value

$$D_2 = \det([D][D]^T) \tag{22}$$

If this D_2 value is near zero, then the full 3D $\bar{\boldsymbol{L}}_r$ vector cannot be achieved. A common example of such a scenario is with the DV thruster configuration where all $\hat{\boldsymbol{g}}_{t_i}$ axes are collinear. Torques about these thrust axes cannot be produced. In this case, the minimum norm solution is adjusted to only match the torques along the control matrix $[C]$ sub-space. Torques being applied outside of $\hat{\boldsymbol{c}}_j$ is not possible as D_2 is essentially zero, indicating the other control axis cannot be controlled with this thruster configuration.

The minimum norm torque solution is now modified to use

$$\bar{\boldsymbol{F}} = ([C][\bar{D}])^T([C][\bar{D}][\bar{D}]^T[C]^T)^{-1}[C]\boldsymbol{L}_r \tag{23}$$

As the thruster configuration cannot produce a general 3D torque, here the $[C]$ matrix must have either 1 or 2 control axes that are achievable with the given thruster configuration.

To set this epsilon parameter, not the definition of the $[D]$ matrix components $\boldsymbol{d}_i = (\boldsymbol{r}_i \times \hat{\boldsymbol{g}}_{t_i})$. Note that $\boldsymbol{r}_i \times \hat{\boldsymbol{g}}_{t_i}$ is a scaled axis along which the ith thruster can produce a torque. The value $\boldsymbol{d}_i$ will be near zero if the dot product of this axis with the current control axis $\hat{\boldsymbol{c}}_j$ is small.

To determine an appropriate ϵ value, let α be the minimum desired angle to avoid the control axis $\hat{\boldsymbol{c}}_j$ and the scaled thruster torque axis $\boldsymbol{r}_i \times \hat{\boldsymbol{g}}_{t_i}$ being orthogonal. If $\bar{r}$ is a mean distance of the thrusters to the spacecraft center of mass, then the d_i values must satisfy

$$\frac{d_i}{\bar{r}} > \cos(90^\circ - \alpha) = \sin\alpha \tag{24}$$

Thus, to estimate a good value of ϵ, the following formula can be used

$$\epsilon \approx d_i^2 = \sin^2\alpha \; \bar{r}^2 \tag{25}$$

For example, if $\bar{r} = 1.3$ meters, and we want α to be at least 1°, then we would set $\epsilon = 0.000515$.

2. $[C]$ **matrix**: The module requires control control axis matrix $[C]$ to be defined. Up to 3 orthogonal control axes can be selected. Let N_c be the number of control axes. The $N_c \times 3$ $[C]$ matrix is then defined as

$$[C] = \begin{bmatrix} \hat{\boldsymbol{c}}_1 \\ \vdots \end{bmatrix} \tag{26}$$

Not that in python the matrix is given in a 1D form by defining `controlAxes_B`. Thus, the $\hat{\boldsymbol{c}}_j$ axes are concatenated to produce the input matrix $[C]$.

3. `thrForceSign Parameter`: Before this module can be run, the parameter `thrForceSign` must be set to either $+1$ (on-pulsing with the ACS configuration) or -1 (off-pulsing with the DV configuration).

4. `use2ndLoop` Flag: If the `thrForceSign` flag is set to $+1$ then an on-pulsing configuration is setup. By default the optional flag `use2ndLoop` is 0 and the algorithm only uses the least-squares fitting loop once. By setting the `use2ndLoop` to $+1$ then the 2nd least squares fitting loop is used during this on-pulsing configuration,.

5. `angErrThresh` **Parameter**: The default value is 0 rad. When the angle between the saturation-clipped torque contribution $\boldsymbol{\tau} = [D]\boldsymbol{F}_{\text{clip}}$ and the reduced allocator target $\bar{\boldsymbol{L}}_r$ exceeds this threshold, commands above their individual thruster limits trigger uniform scaling of $\boldsymbol{F}$. In off-pulsing mode, $\boldsymbol{L}_r$ includes nominal-thrust compensation as defined in section 1.4. To allow an error of α in this angle before scaling, set `angErrThresh` to α in radians. To disable scaling, set the threshold to a value larger than π rad (180°).

REFERENCES

[1] Hanspeter Schaub and John L. Junkins. *Analytical Mechanics of Space Systems*. AIAA Education Series, Reston, VA, 3rd edition, 2014.